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Quiz 1b

1.19, 12

1. Find

- a. $\lg(32)$ 5
 b. $\lg(1/8)$ -3

2 Determine the Θ class of each of the following

- a) $f(n) = 100n^5 + 5n + 1$ n^5
 b) $f(n) = n + \sqrt{n}$ linear
 c) $f(n) = n + 1.001^n$ exponential

3 Is

- a) $n \in O(\lg n)$ no
 b) $n \in O(n \lg n)$ yes
 c) $n \lg n \in O(n^2)$ yes
 d) $n^2 \in O(n \lg n)$ no
 e) $n \in O(n)$ yes
 f) $2^n \in O(3^n)$ yes
 g) $3^n \in O(2^n)$ no
 h) $2^n + n^{100} \in \Theta(2^n)$ yes

4. Order the Θ classes from lowest to highest of the following

3. $F_1(n) = n + 1$
 6. $F_2(n) = 2^n$
 7. $F_3(n) = 3^n$
 2. $F_4(n) = \sqrt{n}$
 1. $F_5(n) = 2$
 4. $F_6(n) = n + n^2$
 5. $F_7(n) = n + n^2 (\lg n)$

5) Find the Θ -Class of the number of steps in terms of N of the following:Input: N a positive integerVariables: I , Sum , K

- a) For $I = 1$ to 3
 $\text{Sum} \leftarrow \text{Sum} + I$ constant

- b) For $I = 1$ to N
 $\text{Sum} \leftarrow \text{Sum} + I$ linear
 $O(n)$

- c) For $I = 1$ to N
 For $K = 0$ to N
 $\text{Sum} \leftarrow \text{Sum} + I$ quadratic
 $O(n^2)$

- d) For $I = 1$ to N
 For $K = 1$ to I
 $\text{Sum} \leftarrow \text{Sum} + I$ quadratic, $\frac{n(n+1)}{2}$
 $O(n^2)^2$

6) Find the Θ -Class of the number of steps in terms of N of the following:Input: N a positive integerVariables: I , Sum , K

- $I \leftarrow 1$
 $\text{Sum} \leftarrow 0$
 While ($I \leq N$)
 For $K = 1$ thru I
 $\text{Sum} \leftarrow \text{Sum} + K$
 $I \leftarrow I + 1$
 quadratic, $\frac{n(n+1)}{2}$
 $O(n^2)^2$